

Iterative decreasing dimension algorithm

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Abstract

In this study, we have given an iterative decreasing dimension method (IDDM) which decreases by one dimension at every step for solution of a system of linear algebraic equations without any pre-processing and an iterative decreasing dimension algorithm (IDDA) which is based on this method. We have also given numerical examples using this algorithm.

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1. Introduction

Studying the solution of the systems of linear algebraic equations

$$AX = f \tag{1}$$

is a classical problem which is important not only in linear algebra but also in applied mathematics. For system (1) where A is an $n \times n$ regular matrix, X and f are n -vectors to be solved for, a decreasing dimension method has already been proposed in [1]. The authors of [1] have partitioned the matrix A into a two by two form:

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix}, \tag{2}$$

where A_{11} is a $k \times k$ matrix, X_1 and u are k -vectors. Two linear systems have been taken from (2):

$$(A_{11} \quad A_{12}) X = u, \tag{3}$$

$$(A_{21} \quad A_{22}) X = v \tag{4}$$

and the homogeneous system of (3) has also been reduced into row-echelon form in [1]. Therefore, all solutions of the homogeneous system of (3) can be written as $X_h = R\alpha$, where $\alpha = (\alpha_1 \alpha_2 \dots \alpha_{n-k})^T$ and $\alpha_1, \alpha_2, \dots, \alpha_{n-k}$ are free variables, R is a matrix which is composed of base vectors of the solutions of the homogeneous system of (3) and

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$\text{rank}(R) = n - k$. Then, by substituting all solutions of (3) into (4), a new linear algebraic system has been written out:

$$A^* \alpha = f^*, \quad (5)$$

where $A^* = \begin{pmatrix} A_{21} & A_{22} \end{pmatrix} R$ is an $(n - k) \times (n - k)$ matrix, $f^* = v - \begin{pmatrix} A_{21} & A_{22} \end{pmatrix} R_0$ is an $(n - k)$ -vector, R_0 is a particular solution of the system (3) [1]. After solving the system (5), there was obtained α and then the solution of the system (1) as $X = R_0 + R\alpha$.

In [2] (therein [3,4]), it has been said that the matrix A^* in (5) is the well known Schur complement of A . Hence the decreasing dimension method proposed in [1] is the same as the well known domain decomposition technique based on a Schur complement type method. Moreover, this method costs more than the standard Schur complement method and does not decrease the dimension of the linear systems. It divides a large system into two smaller systems to be solved separately.

In this study, we have made an improvement by modifying the method of [1]. Without any pre-processing, the modified method decreases the dimension of the linear systems, by one order in every step.

2. Iterative decreasing dimension method (IDDM)

2.1. Symbols

Before we give the method, let us give some symbols and explanations used in the procedure.

- k : $k = 1(1)n, (k = 1, 2, \dots, n)$, iteration number
- m : $m = n - k + 1$, dimension of reduced matrix
- $A^{(k)}$: $m \times m$ – reduced coefficient matrix
- $X^{(k)}$: m – solution vector of reduced system
- $f^{(k)}$: m – right side vector of reduced system
- $A_1^{(k)}$: $1 \times m$ – matrix composed of first row vector of matrix $A^{(k)}$
- $A_2^{(k)}$: $(m - 1) \times m$ – matrix composed remain line vector of matrix $A^{(k)}$
- $u^{(k)}$: 1 – vector composed of first element of vector $f^{(k)}$
- $v^{(k)}$: $(m - 1)$ – vector composed of remain element of vector $f^{(k)}$
- $a_{ij}^{(k)}$: (i, j) element of matrix $A^{(k)}$
- $x_i^{(k)}$: i th element of vector $X^{(k)}$
- $f_i^{(k)}$: i th element of vector $f^{(k)}$
- $X_0^{(k)}$: m – special solution vector of $A_1^{(k)} X^{(k)} = u^{(k)}$
- $R^{(k)}$: $m \times (m - 1)$ – base matrix of solution space of $A_1^{(k)} X^{(k)} = 0$.

2.2. Method

Let us consider a system of linear algebraic equations:

$$AX = A^{(1)} X^{(1)} = f^{(1)} = f; A - \text{regular} \quad (6)$$

where A is an $n \times n$ matrix, X and f are n -vectors. We divide the given system into

$$A_1^{(1)} X^{(1)} = u^{(1)} \quad (7)$$

$$A_2^{(1)} X^{(1)} = v^{(1)} \quad (8)$$

systems by partitioning as follows: $A^{(1)} = A_1^{(1)} \oplus A_2^{(1)}$ and $f^{(1)} = u^{(1)} \oplus v^{(1)}$ where $f = u \oplus v = \begin{pmatrix} u & v \end{pmatrix}^T$.

$X_0^{(1)}$ can be chosen as

$$X_0^{(1)} = \begin{pmatrix} 0 & \dots & 0 & \frac{f_1^{(1)}}{a_{1s}^{(1)}} & 0 & \dots & 0 \end{pmatrix}^T \quad (9)$$

which is a special solution of (7). In (9), $a_{1s}^{(1)} \neq 0$ ($1 \leq s \leq n$), which is the first non-zero element of matrix $A_1^{(1)}$, and then $X_h^{(1)}$, a homogeneous solution of (7), is found to be

$$X_h^{(1)} = R^{(1)} X^{(2)}$$

where $X^{(2)}$ is an $(n-1)$ -vector composed of x_i -parametric variables for $i = 1(1)n, i \neq s$, and $R^{(1)}$ is a matrix which is composed of the base vectors of this solution space:

$$R^{(1)} = \begin{cases} \begin{pmatrix} r_{1 \times (n-1)}^{(1)} \\ I_{(n-1) \times (n-1)} \end{pmatrix} & s = 1, \\ \begin{pmatrix} I_{(s-1) \times (s-1)} & 0_{(s-1) \times (n-s)} \\ 0_{1 \times (s-1)} & r_{1 \times (n-s)}^{(1)} \\ 0_{(n-s) \times (s-1)} & I_{(n-s) \times (n-s)} \end{pmatrix} & s = 2(1)n-1, \\ \begin{pmatrix} I_{(n-1) \times (n-1)} \\ 0_{1 \times (n-1)} \end{pmatrix} & s = n \end{cases}$$

where $r_{1 \times (n-s)}^{(1)} = (r_{1 \times (s+1)}^{(1)} \ r_{1 \times (s+2)}^{(1)} \ \dots \ r_{1 \times n}^{(1)})$; $r_{1j}^{(1)} = -\frac{a_{1j}^{(1)}}{a_{1s}^{(1)}}$, $j = s+1(1)n$. The general solution of (7) is achieved as $X^{(1)} = X_0^{(1)} + R^{(1)} X^{(2)}$, where $X_0^{(1)}$ is an n -vector, $R^{(1)}$ is an $n \times (n-1)$ matrix.

By substituting the $X^{(1)}$ solution into (7), we have a new linear algebraic system in $(n-1)$ dimensions as follows:

$$A^{(2)} X^{(2)} = f^{(2)}; \text{ the system in } (n-1) \text{ dimension,} \quad (10)$$

where $A^{(2)} = A_2^{(1)} R^{(1)}$ and $f^{(2)} = v^{(1)} - A_2^{(1)} X_0^{(1)}$.

Let us apply the steps followed up to here to the system (10). The systems followed by each other can be written as

$$A^{(l)} X^{(l)} = f^{(l)}, \quad l = 2(1)n \quad (11)$$

where $A^{(l)} = A_2^{(l-1)} R^{(l-1)}$ and $f^{(l)} = v^{(l-1)} - A_2^{(l-1)} X_0^{(l-1)}$. Finally, a linear algebraic system in order 1 is reached. Thus, the solution of the system (6) is obtained by substituting the solution $X^{(l+1)}$ into $X^{(l)}$.

Theorem 1. The solution of linear system (6) is

$$X = X^{(1)} = \sum_{i=1}^n \left(\prod_{j=1}^{i-1} R^{(j)} \right) X_0^{(i)},$$

where $\prod_{j=1}^{i-1} R^{(j)} = \begin{cases} R^{(1)} R^{(2)} \dots R^{(i-1)} & i > 1, \\ I & i = 1 \end{cases}$ is defined.

Proof. The general solutions of (7) and (10) are

$$X^{(1)} = X_0^{(1)} + R^{(1)} X^{(2)}$$

and

$$X^{(2)} = X_0^{(2)} + R^{(2)} X^{(3)}$$

respectively. Thus, the general solutions of (11) are

$$X^{(l)} = X_0^{(l)} + R^{(l)} X^{(l+1)}, \quad l = 2(1)n-1,$$

and for $l = n$ the general solution of the system

$$A^{(n)} X^{(n)} = f^{(n)}$$

is obtained as

$$X^{(n)} = X_0^{(n)} = \begin{pmatrix} \frac{f_1^{(n)}}{a_{11}^{(n)}} \end{pmatrix},$$

where

$$A^{(n)} = A_2^{(n-1)} R^{(n-1)} = \begin{pmatrix} a_{11}^{(n)} \end{pmatrix}, \quad f^{(n)} = v^{(n-1)} - A_2^{(n-1)} X_0^{(n-1)} = (f_1^{(n)}).$$

Then, substitution of the $X^{(l+1)}$ solution into $X^{(l)}$ for $l = n - 1(-1)1$ gives the solution of the given system (6) as

$$X = X^{(1)} = \sum_{i=1}^n \left(\prod_{j=1}^{i-1} R^{(j)} \right) X_0^{(i)}. \quad \square$$

Corollary 2. The solution of linear system $A^{(k)} X^{(k)} = f^{(k)}$ ($1 \leq k \leq n$) is

$$X = X^{(k)} = \sum_{i=k}^n \left(\prod_{j=k}^{i-1} R^{(j)} \right) X_0^{(i)}.$$

Proof. From the proof of Theorem 1, it is clear. $\square$

3. Iterative decreasing dimension algorithm (IDDA)

Input. A , regular matrix; f , vector; n , order of A .

Step 1. Get $k = 1(1)n - 1$, calculate $m = n - k + 1$; $i, j = 1(1)m$

$$1.1. A^{(k)} = \begin{pmatrix} A_1^{(k)} \\ A_2^{(k)} \end{pmatrix}; \quad A_1^{(k)} = (a_{1j}^{(k)})_{j=1(1)m}, \quad A_2^{(k)} = (a_{ij}^{(k)})_{i=2(1)m}, \quad A^{(1)} = A$$

$$1.2. f^{(k)} = \begin{pmatrix} u^{(k)} \\ v^{(k)} \end{pmatrix}; \quad u^{(k)} = (f_1^{(k)}), \quad v^{(k)} = (f_i^{(k)})_{i=2(1)m}, \quad f^{(1)} = f.$$

Step 2. Control if $a_{1j}^{(k)} \neq 0$ for $j = 1(1)m$; let $a_{1s}^{(k)} \neq 0$ be first element.

2.1. Calculate the vector $X_0^{(k)}$;

$$X_0^{(k)} = \begin{pmatrix} 0 & \dots & 0 & \frac{f_1^{(k)}}{a_{1s}^{(k)}} & 0 & \dots & 0 \end{pmatrix}^T.$$

2.2. Calculate the matrix $R^{(k)}$;

$$R^{(k)} = \begin{cases} \begin{pmatrix} r_{1 \times (m-1)}^{(k)} \\ I_{(m-1) \times (m-1)} \end{pmatrix} & s = 1, \\ \begin{pmatrix} I_{(s-1) \times (s-1)} & 0_{(s-1) \times (m-s)} \\ 0_{1 \times (s-1)} & r_{1 \times (m-s)}^{(k)} \\ 0_{(m-s) \times (s-1)} & I_{(m-s) \times (m-s)} \end{pmatrix} & s = 2(1)m - 1, \\ \begin{pmatrix} I_{(m-1)1 \times (m-1)} \\ 0_{1 \times (m-1)} \end{pmatrix} & s = m \end{cases}$$

where

$$r^{(k)} = \begin{pmatrix} r_{1 \times (s+1)}^{(k)} & r_{1 \times (s+2)}^{(k)} & \cdots & r_{1 \times m}^{(k)} \end{pmatrix}; \quad r_{1j}^{(k)} = -\frac{a_{1j}^{(k)}}{a_{1s}^{(k)}}, \quad j = s+1(1)m.$$

2.3. Calculate $A^{(k)}$ and $f^{(k)}$;

$$A^{(k)} = A_2^{(k-1)} R^{(k-1)}; \quad f^{(k)} = v^{(k-1)} - A_2^{(k-1)} X_0^{(k-1)}, \quad k = 2(1)n-1.$$

Step 3. Get $k = n$;

3.1. Calculate $A^{(n)} = A_2^{(n-1)} R^{(n-1)}$; $f^{(n)} = v^{(n-1)} - A_2^{(n-1)} X_0^{(n-1)} = (f_1^{(n)})$.

3.2. Calculate $X_0^{(n)} = \begin{pmatrix} f_1^{(n)} \\ a_{11}^{(n)} \end{pmatrix}$.

Output. Solution

$$X = \sum_{i=1}^n \left(\prod_{j=1}^{i-1} R^{(j)} \right) X_0^{(i)}.$$

4. Numerical examples

Example 1.

$$A = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 2 & -1 \\ 1 & 0 & 1 \end{pmatrix}, \quad f = \begin{pmatrix} 7 \\ 2 \\ 4 \end{pmatrix}, \quad n = 3$$

1. For $k = 1$,

$$A^{(1)} = A, \quad f^{(1)} = f, \quad X_0^{(1)} = (0 \quad 3.5 \quad 0)^T, \quad R^{(1)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0.5 & 1 \end{pmatrix}^T$$

2. For $k = 2$,

$$A^{(2)} = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}, \quad f^{(2)} = (-5 \quad 4)^T, \quad X_0^{(2)} = (-5 \quad 0)^T, \quad R^{(2)} = (2 \quad 1)^T$$

3. For $k = 3$, $A^{(3)} = (3)$, $f^{(3)} = (9)$, $X_0^{(3)} = (3)$

Output. Solution

$$X = \sum_{i=1}^3 \left(\prod_{j=1}^{i-1} R^{(j)} \right) X_0^{(i)} = (1 \quad 2 \quad 3)^T.$$

Example 2.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 2 \\ 3 \\ 1 \\ -1 \end{pmatrix}, \quad n = 4$$

1. For $k = 1$,

$$A^{(1)} = A, \quad f^{(1)} = f, \quad X_0^{(1)} = (0 \quad 2 \quad 0 \quad 0)^T, \quad R^{(1)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}^T$$

2. For $k = 2$,

$$A^{(2)} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad f^{(2)} = (3 \quad 1 \quad -1)^T, \quad X_0^{(2)} = (0 \quad 0 \quad 3)^T, \quad R^{(2)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}^T$$

3. For $k = 3$,

$$A^{(3)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad f^{(3)} = (1 \quad -1)^T, \quad X_0^{(3)} = (1 \quad 0)^T, \quad R^{(3)} = (0 \quad 1)^T$$

4. For $k = 4$,

$$A^{(4)} = (1), \quad f^{(4)} = (-1), \quad X_0^{(4)} = (-1)$$

Output. Solution

$$X = \sum_{i=1}^4 \left(\prod_{j=1}^{i-1} R^{(j)} \right) X_0^{(i)} = (1 \quad 2 \quad -1 \quad 3)^T.$$

Remark. We have seen that the some linear systems cannot be directly solved using the method in [1]. For example, the matrix A in [Example 2](#) cannot be partitioned as in (2) such that $\text{rank}(A_{11}) = k < 4$ without any pre-processing.

5. Conclusion

IDDM produces a special $X_0^{(k)}$ solution and $R^{(k)}$ matrix by reducing the dimension of a given system of linear algebraic equations by one dimension at every step. It obtains the solution depending on special $X_0^{(k)}$ solutions and $R^{(k)}$ matrices. For DDM to be applicable, A_{11} ($k < n$), a regular matrix, is needed but IDDM does not need it. For example, DDM cannot be directly applied to the A matrix in [Example 2](#) to be solved. For DDM to be applied, the $(A_{11} \quad A_{12})$ matrix must be in row-echelon form or reduced into row-echelon form. IDDM does not need this process. IDDM can be applied manually or using a computer.

There is an argument that IDDA is based on IDDM. The output of IDDA is the solution of the given system.

References

- [1] H. Wang, J. Jiang, Solution of the system of linear algebraic equations by decreasing dimension, *Appl. Math. Comput.* 109 (2000) 51–57.
- [2] J. Zhang, Comments on ‘Solution of the system of linear algebraic equations by decreasing dimension’, *Appl. Math. Comput.* 128 (2002) 95–98.
- [3] G.H. Golub, J.M. Ortega, *Scientific Computing: An Introduction with Parallel Computing*, Academic Press, Boston, MA, 1993.
- [4] G.H. Golub, C.F. van Loan, *Matrix Computations*, The Johns Hopkins University Press, Baltimore, MD, 1983.